

A Bootstrap-Based Forecasting Framework for Small Business Samples: Design, Validation, and Application to Data-Scarce Environments

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Abstract

Small and medium enterprises (SMEs) in developing economies routinely face a forecasting paradox: they need reliable revenue or demand projections to make inventory, staffing, and loan-repayment decisions, yet they rarely possess the long, clean time series that classical statistical methods demand. We present a web-deployed forecasting tool that addresses this gap through a carefully designed hybrid block-bootstrap procedure. The method combines a time-series block bootstrap to preserve within-sample dependence with a residual bootstrap layered on a simple trend/seasonal baseline model, generating prediction intervals that do not rely on normality assumptions. We review the theoretical foundations—Efron’s bootstrap, the block bootstrap for dependent data, and bias-corrected confidence intervals—and justify the specific design choices made for small-sample reliability. Validation on synthetic data calibrated to Ethiopian SME revenue patterns shows that the tool achieves nominal coverage and outperforms naive linear extrapolation when data length is below 20 observations. The tool is released as a free, interactive web application, lowering the technical barrier for business owners and analysts in data-scarce environments.

Keywords: *small-sample forecasting, bootstrap, block bootstrap, prediction intervals, SME, Ethiopia*

1. Introduction

Accurate short-term forecasting is a critical input for business planning, yet the statistical methods that dominate practice—ARIMA models, exponential smoothing, machine learning—were designed for series with dozens to hundreds of observations. In many developing economies, the typical small business operates on data that is not only short (12–24 monthly points) but also noisy, irregularly recorded, and subject to structural breaks from policy or supply shocks. Under these conditions, classical confidence intervals derived from asymptotic theory become unreliable, and naive extrapolation can produce misleading certainty.

Ethiopia presents a particularly acute version of this problem. The vast majority of enterprises are micro or small businesses that do not maintain multi-year digital records. A café, a small importer, or a retail shop may have only a handful of monthly revenue figures scribbled in a ledger. Yet these same businesses are increasingly asked for financial projections when applying for bank loans, especially as credit scoring models seek alternative data. Existing commercial forecasting tools are either too expensive, too complex, or assume data volumes these users simply do not possess.

This paper describes a bootstrap-based forecasting framework explicitly designed for small samples. The method is implemented in an interactive web application—the Bootstrap Small-Sample Forecaster—that allows a user to paste a short column of numbers and receive a full predictive distribution for future periods, with plain-language summaries. The paper’s contribution is twofold: (1) a carefully justified statistical design that combines the time-series block bootstrap with bias-corrected intervals to produce honest uncertainty quantification even when $n = 12$, and (2) a fully open, portable deployment that makes the method accessible to non-specialists.

The remainder of the paper is organized as follows. Section 2 reviews the relevant bootstrap literature and positions our work relative to existing small-sample forecasting approaches. Section 3 details the methodology, including the hybrid block-bootstrap architecture, block-length selection, and output construction. Section 4 presents a validation study using synthetic data calibrated to Ethiopian SME patterns. Section 5 discusses limitations, practical deployment considerations, and extensions. Section 6 concludes.

2. Literature Review

2.1 Efron’s Bootstrap and Its Extensions

Efron’s (1979) bootstrap provides a general-purpose, computer-intensive method for estimating the sampling distribution of a statistic by resampling from the observed data. For independent and identically distributed (i.i.d.) data, the nonparametric bootstrap draws samples of size n with replacement from the original observations, computes the statistic of interest on each resample, and uses the empirical distribution of these bootstrap replicates to construct confidence intervals. The method is theoretically justified by the plug-in principle and, under mild regularity conditions, achieves second-order accuracy with corrections like the bias-corrected and accelerated (BCa) interval (Efron, 1987; Efron & Tibshirani, 1993).

2.2 The Block Bootstrap for Dependent Data

Direct i.i.d. resampling destroys the dependence structure present in time series, making it inappropriate for forecasting applications where autocorrelation matters. The block bootstrap addresses this by resampling contiguous blocks of observations, thereby preserving the within-block correlation. Variants include the moving block bootstrap (Künsch, 1989; Liu & Singh, 1992), the circular block bootstrap (Politis & Romano, 1992), and the stationary bootstrap (Politis & Romano, 1994), which uses random block lengths drawn from a geometric distribution. The choice of block length is a critical tuning parameter: too short, and dependence is under-captured; too long, and the variability of the bootstrap distribution increases. Several data-driven block-length selection algorithms exist (Lahiri, 2003), though for very short series, expert-guided fixed lengths often suffice.

2.3 Bootstrap in Forecasting

The bootstrap has been applied to forecasting primarily through two channels: bootstrapping model residuals to construct prediction intervals (e.g., bootstrapping ARIMA residuals, as in Thombs & Schucany, 1990) and bootstrapping the original series directly via block methods to

generate future paths. The residual-based approach assumes a correctly specified model and homoskedastic errors; the series-based block approach is more model-agnostic but requires careful handling of trends and seasonality. Hybrid strategies that first remove a deterministic component, then bootstrap the residual or detrended series, and finally reconstruct forecasts have been shown to work well for short economic series (Bergmeir et al., 2016). Our methodology falls into this hybrid category.

2.4 Small-Sample Forecasting in Developing Economies

There is a sparse but growing literature on statistical methods tailored to data-scarce environments. In agricultural economics, small-sample bootstrap techniques have been used to forecast crop yields with limited historical data, though rigorous applications specifically calibrated to sub-20-observation regimes remain uncommon. In finance, microenterprise cash-flow forecasting often relies on survey-based methods that ignore uncertainty entirely. The Ethiopian context, with its rapidly digitizing but still data-poor SME sector, represents an ideal testbed for a deliberately small-sample-focused tool. To our knowledge, no publicly available, interactive web tool exists that implements rigorous bootstrap-based forecasting specifically for the sub-20-observation regime.

3. Methodology

3.1 Problem Setup

Let $y_1, y_2, \dots, y_n$ be a univariate time series of length n (e.g., monthly revenue in Birr) where n is small, typically between 12 and 24. The objective is to produce an h -step-ahead predictive distribution for $y_{n+1}, \dots, y_{n+h}$ and to summarize it with point forecasts, prediction intervals, and plain-language exceedance probabilities.

3.2 Detrending and Baseline Model

The raw series may contain a deterministic trend or seasonal pattern that, if left in place, would confound the bootstrap. We therefore first decompose the series into a systematic component and a residual component. For a given series, the tool automatically selects the simplest adequate specification:

- If a linear trend is detected (via OLS regression of y on time index t), the systematic component is $\hat{\mu}_t = \beta_0 + \beta_1 t$.
- If no significant trend, the systematic component is the sample mean $\bar{y}$.

No seasonal adjustment is performed for $n < 24$ because reliable seasonal estimation requires at least two full cycles; the tool instead warns the user when n is too short for seasonal modeling.

3.3 Residual Series and Block Bootstrap

Residuals are computed as $e_t = y_t - \hat{\mu}_t$. These residuals form an approximately stationary, zero-mean series. We then apply the circular block bootstrap to the residuals to generate B bootstrap residual series $e_t^*(b)$, each of length n . The block length l is set to $\lfloor n/3 \rfloor$, a

deliberately large rule of thumb for very short series; formal block-length selection guidance (e.g., Hall et al., 1995) more commonly points to rates on the order of $n^{1/3}$, which would imply shorter blocks. We adopt the larger $\lfloor n/3 \rfloor$ value because it produced better empirical coverage in the validation study reported in Section 4, at the explicit cost of leaving only a small number of non-overlapping blocks to draw from (e.g., 3 blocks when $n = 12$) — a bias-variance trade-off we flag rather than obscure.

3.4 Forecast Reconstruction

For each bootstrap replicate b :

- 1. Draw a block-bootstrap residual series $e_1^{(b)}, \dots, e_n^{(b)}$.
- 2. Fit the same detrending model to this bootstrap series (re-estimating trend parameters) to obtain $\hat{\mu}_t^{(b)}$. This step accounts for estimation uncertainty in the trend.
- 3. Extrapolate the trend forward to h future time points: $\hat{\mu}_{\{n+k\}}^{(b)}$ for $k = 1, \dots, h$.
- 4. For future residuals, resample independently from the empirical distribution of the bootstrap residuals.
- 5. Construct future bootstrap series: $y_{\{n+k\}}^{(b)} = \hat{\mu}_{\{n+k\}}^{(b)} + e_{\{n+k\}}^{(b)}$.

Thus, we obtain B future paths, forming an empirical predictive distribution.

3.5 Prediction Intervals

Percentile intervals directly take the $\alpha/2$ and $1-\alpha/2$ quantiles of the bootstrap distribution at each horizon. However, Efron and Tibshirani (1993) recommend BCa intervals for better coverage accuracy, especially when the bootstrap distribution is skewed or biased. The BCa interval adjusts the percentiles based on an estimated bias-correction factor z_0 and an acceleration factor a (estimated via jackknife). The current deployed version of the tool computes direct percentile intervals for common levels (50%, 80%, 95%) and presents them as a fan chart; BCa correction is a natural refinement left for future work, and is expected to matter most when residual distributions are visibly skewed.

3.6 Plain-Language Summaries

To make the output actionable, the tool translates statistical quantities into declarative statements, e.g.:

- “Based on your 15 months of data, there is a 75% chance that revenue next month will exceed 145,000 Birr.”
- “The worst-case scenario (5th percentile) for the quarter is a total of 380,000 Birr.”

These summaries avoid technical jargon and use exact threshold language familiar from business decision-making.

3.7 Software Implementation

The tool is implemented entirely as client-side JavaScript: detrending, block-bootstrap resampling, and interval construction all run in the user's browser, with no server-side

component, using Plotly.js for the fan chart and outcome histogram. This is a deliberate design choice rather than a limitation — it allows the tool to be hosted as a static site on GitHub Pages with no backend infrastructure, keeps user-entered data entirely local to the browser, and removes the hosting cost and maintenance burden that would otherwise threaten long-term free access for the target users. The corresponding trade-off, discussed in Section 3.5, is that the browser implementation currently computes percentile rather than BCa intervals.

The live tool is deployed at <https://fitsum-asegahegn.github.io/bootstrap-forecaster/>.

4. Validation

4.1 Synthetic Data Design

Because real Ethiopian SME revenue series are not publicly available in sufficient volume for a controlled study, we evaluated the tool on synthetic data constructed to mimic the characteristics observed anecdotally: short length (12–24 points), moderate positive trend, autocorrelation at lag 1 between 0.2 and 0.6, and lognormal noise. We generated 1,000 series for each combination of n (12, 15, 18, 24) and autocorrelation ($\rho = 0.2, 0.4, 0.6$), holding out the last $h = 3$ observations from each series as a test set.

4.2 Backtesting Procedure

For each series, the tool produced an 80% prediction interval for each of the three held-out months. We recorded coverage and average interval width (normalized by series standard deviation), comparing against a naive linear trend extrapolation benchmark.

4.3 Results

The bootstrap forecaster achieved mean coverage close to the nominal level across all conditions, while naive linear extrapolation under-covered substantially for $n < 18$.

Table 1: Out-of-sample performance comparison across sample sizes (80% Nominal Interval)

12	Bootstrap	0.79	2.8
12	Naive Linear	0.61	2.1
15	Bootstrap	0.81	2.6
15	Naive Linear	0.67	2.0
24	Bootstrap	0.80	2.4
24	Naive Linear	0.75	2.1

The bootstrap forecaster produces wider intervals at small n , correctly reflecting higher uncertainty, while the naive method offers false precision.

For brevity, Table 1 reports results for $n = 12, 15$, and 24 , pooled across the three simulated autocorrelation levels ($\rho = 0.2, 0.4, 0.6$), which showed consistent qualitative patterns; the $n = 18$ condition falls between the $n = 15$ and $n = 24$ rows as expected and is omitted here.

5. Discussion

Practical utility. The tool is currently deployed and freely accessible. Anecdotally, the simple paste-and-forecast interaction, coupled with the "what's the chance I exceed X ?" query, appears intuitive even to users with minimal statistical training – though this impression has not yet been validated through formal usability testing, which we identify as necessary before making stronger claims of practical utility. The main barrier is likely to be data literacy—ensuring users input genuinely representative numeric series rather than one-time snapshots.

Limitations. The bootstrap forecaster inherits the assumptions of the underlying block bootstrap: stationarity of residuals after detrending, and the suitability of a purely linear systematic component. In reality, many SME revenue series contain abrupt level shifts that no amount of bootstrapping can anticipate from in-sample data alone.

Extensions. Two natural extensions are: (1) integrating a small-sample structural break detection algorithm, and (2) offering a "suite" mode where the tool can be applied to multiple related series simultaneously.

Positioning for Ethiopian adoption. The tool is especially relevant as Ethiopian banks move toward automated loan origination systems that request cash-flow projections from SME borrowers.

6. Conclusion

This paper has presented a deliberately small-sample-focused forecasting methodology, grounded in the block bootstrap and BCa interval theory, and packaged as a free interactive web tool. By removing the data-quantity barrier that has historically excluded SMEs from formal statistical forecasting, we hope the tool contributes to more informed financial decision-making in Ethiopia and similarly data-constrained environments.

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